

CST4714: Database Administration

Course Introduction & Overview

Computer Systems Technology Department, City Tech

Course Topics & Objectives

Week 1: DBA tasks, roles & review of design

Week 2: Database architecture & memory/transactions

Week 3: Tablespaces & storage management

Week 4: Schema management & SQL object types

Week 5–6: Concurrency, locking & installing/upgrading

Weeks 7–10: DB creation, user management & security

Weeks 11–14: Performance tuning, backup/recovery & advanced topics

Final Exam & project submission

DBMS Fundamentals & Role of the DBA

Why DBMS?

File-based systems suffer from data redundancy & isolation
Difficult to enforce consistency across multiple files
Hard to access and update data; no centralized catalogue
DBMS provides concurrency control, recovery & security

Duties of a DBA

Plan: analyze requirements & choose technologies
Build: install DBMS & create structures
Tune: optimise queries, indexes, memory
Troubleshoot: diagnose and resolve issues
Secure: manage users & privileges, audit usage
Monitor: observe metrics & anticipate problems



Storage

Query Processor

Concurrency

Recovery

Data Models & Architecture

Levels of Abstraction

External (view) level: tailored user views
Conceptual (logical) level: overall schema
Physical (internal) level: storage details
Data independence between levels

Architectures

Centralised vs client–server vs distributed
Three-tier model: client, app server, DB server
Parallel DBs for performance & availability
Understand memory, processes & storage



Relational Model Essentials

Data organised as relations (tables)

Relation = set of tuples (rows) and attributes (columns)

Domains define the legal values of attributes

Schema vs instance: design vs current contents

Keys uniquely identify tuples; foreign keys enforce relationships

studentId	name	major
1001	Alice	CST
1002	Bob	CST
1003	Carol	CET

Introduction to SQL

SQL Components

Data Definition Language (DDL)

Data Manipulation Language (DML)

Data Control Language (DCL)

Declarative syntax & portability

```
-- Example SELECT query
SELECT name, major
FROM student
WHERE major = 'CST';
```

Entity–Relationship Modeling

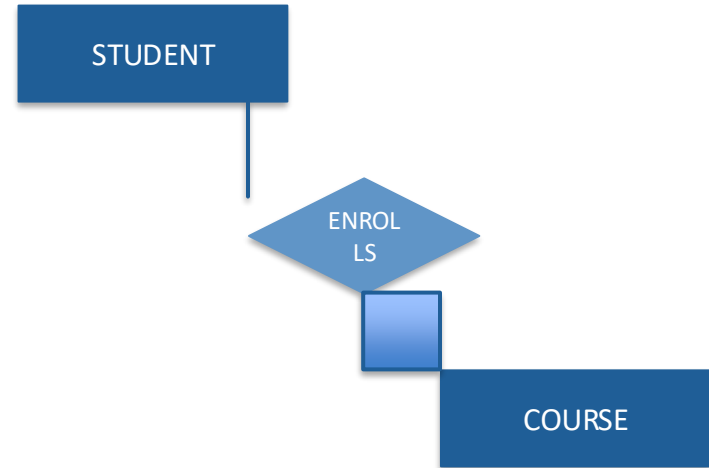
Concepts

Entities: objects with attributes

Relationships: associations between entities

Cardinality: one-to-one, one-to-many, many-to-many

ER diagrams aid communication & design



Normalization & Database Design

Why Normalize?

Reduce redundancy & prevent anomalies

1NF: atomic values & unique rows

2NF: full dependency on the key

3NF/BCNF: remove transitive dependencies

Balance normalisation against performance

Normal Form	Description	Goal
1NF	Eliminate repeating groups	Atomic attributes
2NF	No partial dependency	Full key dependence
3NF/BCNF	No transitive dependency	Dependency preservation

Course Outline & Logistics

Weekly Highlights

- 1: DBA role & design review
- 2: Architecture, memory & transactions
- 3: Tablespaces & storage
- 4: Schema management
- 5: Concurrency & locking
- 6: Installation & upgrades
- 7: Midterm review/exam
- 8: Creating databases & parameter files
- 9: User management & security
- 10: Data loading & transformation
- 11: Query optimisation
- 12: Performance tuning & memory
- 13: Backup & recovery
- 14: Advanced topics; distributed & warehouse
- 15: Final exam & project

Assessment	Weight
Midterm	20%
Final	30%
Quizzes	20%
Case Project	25%
Homework	5%

Wrap-Up & Next Steps

Summary & Looking Ahead

Reviewed course scope, objectives & logistics

Introduced DBMS benefits & DBA responsibilities

Explored abstraction levels & architectures

Discussed relational model, SQL, ER & normalization